
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, exact weight analysis, and causal representation tests

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a 20.1M-
4 parameter vision-language-action policy built only from bilinear, linear, and rational
5 learned operations. Its weights can therefore be reorganized into explicit interaction
6 tensors rather than treated only as an opaque checkpoint. A mechanical audit and
7 numerical reconstructions verify that the deployed policy preserves this form.
8 Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches $93.7\% \pm$
9 0.8% over three seeds under the published protocol. In a separate ViT instantiation,
10 we reconstruct every attention module in three checkpoints with worst relative error
11 2.56×10^{-7} , and exact weight-derived subspaces beat equal-rank random controls
12 in all 36 primary block-head tests. Across three checkpoints, an exploratory rank
13 screen finds that retaining 96 of 384 causally ranked visual dimensions succeeds in
14 50/60 matched trials, compared with 52/60 for the full representation. Three equal-
15 rank random controls reach only 1/60, 3/60, and 3/60. The selected subspace
16 retains 49 of the 52 full-policy successes. On disjoint cache samples, it has
17 $33.5\text{--}110.6\times$ lower reconstruction error than the median across three random
18 projectors, computed separately by action group. These exact and causal analyses
19 are complementary. They do not yet map the exact attention factors directly to the
20 causal visual basis. The result is a capable and structurally inspectable specialist
21 family with a compact exploratory policy bottleneck.

22 1 Introduction

23 Modern robot policies can achieve high success while giving little indication of why an action was
24 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
25 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
26 activations, yet a correlation is not automatically a mechanism or a safe editing target.

27 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
28 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
29 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
30 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
31 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
32 tensor-decomposable policy.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis and causal representation tests?

We test capability and structural analysis in two fully rational encoder instantiations. The convolutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact attention audit and causal representation screen.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$ over three seeds, and a three-checkpoint ensemble reaches 96.2%.
3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor.
5. In a separate exploratory rank screen, a selected 96-dimensional visual subspace retains 49 of 52 full-policy successes across three checkpoints. Three equal-rank random controls reach only 1/60, 3/60, and 3/60.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible and supports separate causal tests of its internal representations.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D [(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

75 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 76 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

77 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 78 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 79 rescaling without leaving the rational tensor representation.

80 2.2 Architecture and training

81 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 82 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 83 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 84 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 85 continuous action dimensions. The complete model has 20,137,352 learned parameters.

86 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 87 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 88 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 89 that verify every requested state was loaded.

90 2.3 Mechanical certificate on the trained checkpoint

91 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 92 point at runtime and reconstruct representative operations from saved weights.

93 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 94 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 95 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
 96 summarized in Appendix Figure 5. It does not materialize one compact polynomial for the full
 97 eight-layer transformer.

98 3 Protocol-aligned closed-loop capability

99 3.1 Evaluation protocol

100 We evaluate all ten LIBERO-Object tasks with:

- 101 • 50 canonical trials per task
- 102 • a 280-step cap
- 103 • three independently trained seeds per architecture
- 104 • 500 rollouts per seed and 1,500 rollouts per architecture

105 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 106 rerun in our codebase, so differences may still include implementation and training choices.

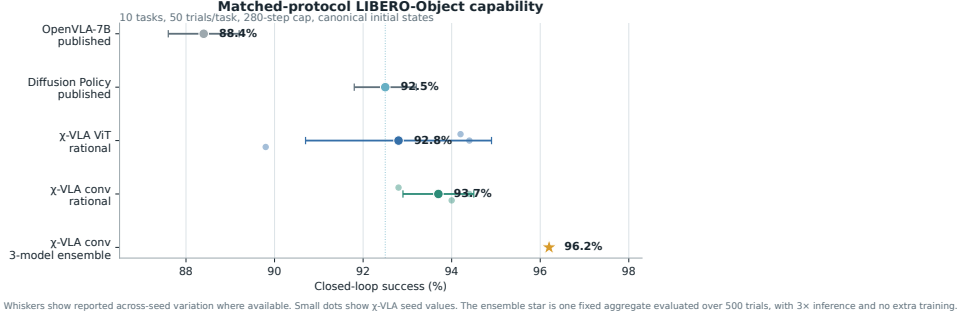


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1M$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the

131 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
132 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

133 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
134 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
135 rank important directions without constructing the full table. It matches brute-force materialization to
136 2.13×10^{-14} at toy scale.

137 4.2 Trained policy result

138 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
139 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
140 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
141 activations are used only afterward to score how well it preserves the layer’s computation.

142 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
143 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
144 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
145 0, 6, and 7.

146 This extends exact weight-only construction through individual attention layers. It is not an exact
147 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
148 rapid degree and representation growth.

149 5 Causal policy structure

150 We next test the policy’s representation with a separate causal analysis.

151 5.1 Causal visual subspace

152 At the visual bond before the joint backbone, we build one shared action-balanced ranking. We form
153 one Jacobian Gram per action coordinate, trace-normalize the seven Grams, and average them. A
154 subspace is a smaller coordinate system inside the model’s 384-dimensional visual representation.
155 Retaining it means zeroing all other directions, then running the actual policy. We compare selected
156 and random equal-width subspaces. Because the ranking uses cached training demonstrations and
157 downstream action gradients, offline reconstruction remains in-cache. Its evaluation samples are
158 disjoint from the samples that estimate the ranking. Closed-loop intervention uses independent
159 canonical simulator states, so it tests a causal bottleneck but is not a weight-only discovery result. The
160 exploratory screen tests ranks 64, 96, 128, and 192 on tasks 4 to 7. Appendix Figure 2 reports the full
161 curve. Rank 96 is the smallest tested width with more than 90% conditional retention while random
162 success remains below 5%. Tasks 8 and 9 are reserved for a frozen confirmation and are not used
163 here. At rank 96, full representations succeed in 17/20, 15/20, and 20/20 trials across checkpoints.
164 Selected rank-96 subspaces reach 17/20, 14/20, and 19/20, and retain 49 of the 52 exact full-policy
165 successes. Their conditional retention is 94.1%, 93.3%, and 95.0%. Three random controls per
166 checkpoint total only 1/60, 3/60, and 3/60. Every checkpoint and random-control comparison
167 has the same selected-over-random direction. Within each fixed checkpoint, the descriptive exact
168 paired tests have $p < 2.5 \times 10^{-4}$. Offline fidelity uses 1,024 discovery samples and 1,024 disjoint
169 evaluation samples per checkpoint. The selected basis has 53.0–110.6 $\times$ lower translation error,
170 37.5–62.1 $\times$ lower rotation error, and 33.5–103.0 $\times$ lower gripper error than the median across three
171 random projectors, computed separately by action group. This repairs sample reuse but remains an
172 in-cache diagnostic. This data-driven full-policy ranking complements the exact layerwise analysis in
173 Section 4.

174 6 Related work and limitations

175 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
176 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
177 weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained
178 VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a

179 policy designed as a rational tensor program. Tensor and polynomial networks provide the broader
180 multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success
181 can coexist with weak language use [9–11].

182 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
183 robotic generalization. Exact reconstruction is certified layer by layer, while the full-policy visual
184 ranking also uses cached activations and action gradients. The capability result uses convolutional
185 checkpoints, while the exact and causal analyses use separate ViT checkpoints. The latter are
186 complementary, not a demonstrated direct mapping. Evaluation uses fixed task instructions and
187 does not establish robust instruction grounding. Within that scope, rational restrictions support
188 strong LIBERO-Object control, exact trained-weight certificates, and an exploratory rank-96 causal
189 bottleneck that retains 49 of 52 full-policy successes across three checkpoints. We do not claim a
190 direct coefficient-edit interface.

191 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
192 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
193 unavailable.

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A Supporting results and decisions

A.1 Rank screen and selectivity

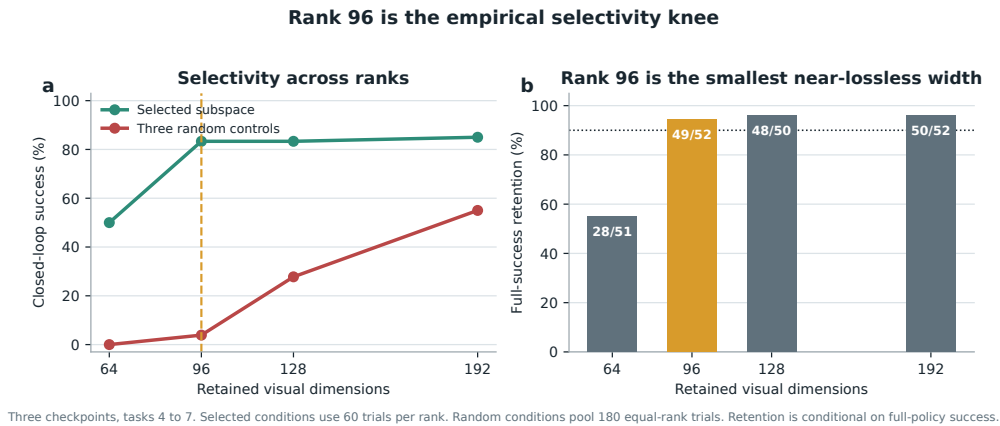


Figure 2: **Rank 96 is the empirical selectivity knee.** Selected subspaces use 60 closed-loop trials per rank across three checkpoints. Random results pool three equal-rank controls per checkpoint, for 180 trials per rank. Each rank has its own paired full-policy condition. Rank 96 is the smallest width above 90% conditional retention, while random success remains below 5%.

221 The selected subspace rises from 30/60 successes at rank 64 to 50/60 at rank 96. Wider ranks
 222 add little selected success, reaching 50/60 and 51/60 at ranks 128 and 192. In contrast, random
 223 success rises from 7/180 at rank 96 to 50/180 and 99/180. Conditional retention is 54.9%, 94.2%,
 224 96.0%, and 96.2% across the four ranks. Thus rank 96 preserves nearly all baseline successes before
 225 equal-width random subspaces become broadly capable.

226 A.2 Causal visual-subspace intervention

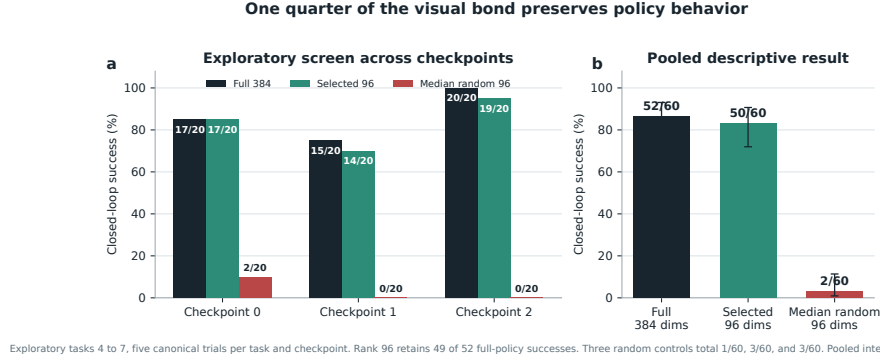


Figure 3: **One quarter of the visual bond preserves policy behavior in the exploratory rank screen.** Each checkpoint uses four tasks and five paired canonical trials per task. Rank 96 retains 49 of 52 full-policy successes. Three equal-rank random controls reach 1/60, 3/60, and 3/60. The pooled interval is descriptive because trials within a checkpoint share learned weights.

227 A.3 Per-task ensemble behavior

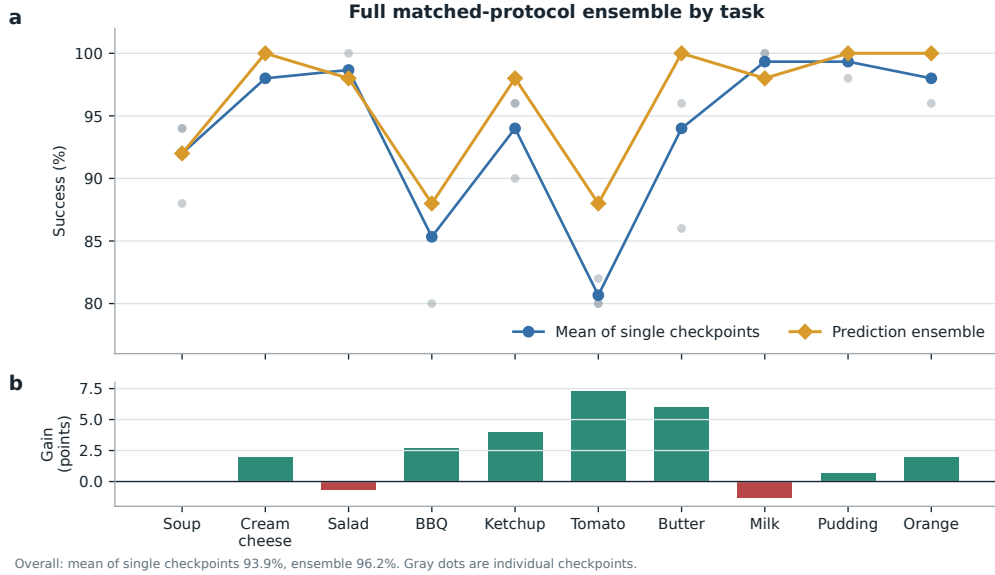


Figure 4: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

228 A.4 Expanded attention screen

229 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 230 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 231 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 232 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 233 tested rank.

234 A.5 Numerical scope of rational conversion

235 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 236 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 237 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 238 establish practical numerical safety under distribution shift.

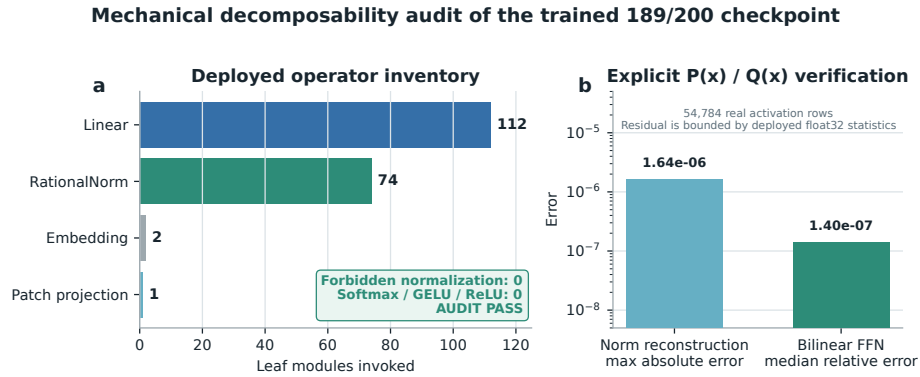


Figure 5: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.